

K-Means Clustering in ML

K-Means clustering : It is one of the most important and commonly used unsupervised machine learning algorithms. It is especially used for understanding the concept of clustering, customer segmentation, grouping similar data points, and exploratory data analysis.

Clustering : Clustering is an unsupervised learning technique to divide data points into groups called clusters. Where, data points within the same cluster are more similar to each other than to data point in other clusters.

Things to consider :

K = number of clusters

Means = cluster center calculated using the mean of data points.

Distance between data points using Euclidean distance.

cluster centroids

Scaling and normalization

Features used for clustering.

K-Means Algorithm Steps

- Step 1: Select the number k to decide the number of clusters.
- Step 2: Select random k points or centroids. (It can be other from the input dataset).
- Step 3: Assign each data points to its nearest centroid, which will form predefined k clusters.
- Step 4: Calculate the ^{mean} ~~variance~~ and place a new centroid of each cluster.
- Step 5: Repeat step 3 and step 4 until the centroids stop changing significantly.

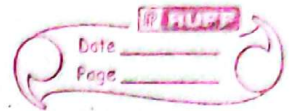
How K-Means clustering works

Ques:

Numerical Data

Example:

Numerical Data



We cluster 12 people by height and weight with $k=2$.

person	Height	Weight
P_1	185	72
P_2	170	56
P_3	168	60
P_4	179	68
P_5	182	72
P_6	188	77
P_7	180	71
P_8	180	70
P_9	183	84
P_{10}	180	88
P_{11}	180	67
P_{12}	177	76

Groups

Group

$$K_1 = P_1 = (185, 72) \quad K_2 = P_2 = (170, 56)$$

Step 1: For point $P_3 (168, 60)$

we use Euclidean distance: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
Distance from P_3 to $K_1 (185, 72)$

$$d_{K_1} = \sqrt{(168 - 185)^2 + (60 - 72)^2} = \sqrt{289 + 144} \\ = \sqrt{433} = 20.81$$

Distance from P_3 to $K_2 (170, 56)$

$$d_{K_2} = \sqrt{(168 - 170)^2 + (60 - 56)^2} \\ = 4.47$$

comparing d_{k_1} and d_{k_2} , minimum distance for point P_3 is d_{k_2} which is 4.47 so, P_3 goes to k_2 group.

$$\text{Group } k_2 = \{P_2, P_3\}$$

update centroid for that since P_3 comes to k_2 group we update centroid of k_2 i.e.

$$k_{2 \text{ new}} = \left(\frac{170 + 168}{2}, \frac{55 + 60}{2} \right) \\ = (169, 58)$$

New,

$$k_1 = (185, 72)$$

$$k_2 = (169, 58)$$

step 2: Point $P_4 (179, 68)$

Distance from P_4 to $k_1 (185, 72)$:

$$d_{k_1} = \sqrt{(179 - 185)^2 + (68 - 72)^2} = 7.21$$

$$d_{k_2} = \sqrt{(179 - 169)^2 + (68 - 58)^2} = 14.14$$

Distance from P_4 to $k_2 (169, 58)$:

$$d_{k_2} = \sqrt{(179 - 169)^2 + (68 - 58)^2} = 14.14$$

Comparing,

minimum distance for point P_4 is k_1 , with 7.21 so, P_4 goes to k_1 group.

$$\text{Group } k_1 = \{P_1, P_4\}$$

update centroid for ~~18~~ k_1 ,

$$k_{1, \text{new}} = \left(\frac{185 + 179}{2}, \frac{72 + 68}{2} \right) = (182, 70)$$

Now,

$$k_1 = (182, 70)$$

$$k_2 = (169, 58)$$

Again, For P_5 and all,

For P_5

$$d_{k_1} = 2.00 \text{ and } d_{k_2} = 19.10$$

So, ~~P_5~~ P_5 comes under k_1 group.

$$\text{new } k_1 = \left(\frac{185 + 179 + 182}{3}, \frac{72 + 68 + 72}{3} \right) = (182, 70)$$

For P_6

$$d_{k_1} = 8.72 \text{ and } d_{k_2} = 26.87$$

So P_6 also comes under k_1 group

$$\text{update } k_1 = (183.5, 72.2)$$

$$k_2 = (169, 58)$$

Step	Point	d_{k_1}	d_{k_2}	Min	Chose to update centroid
5	$P_7(180, 71)$	3.72	17.03	d_{k_1}	$k_1 = (182.8, 72.0)$
6	$P_8(180, 70)$	3.44	16.28	"	$k_1 = (182.3, 71.7)$
7	$P_9(183, 84)$	12.35	29.58	"	$k_1 = (182.4, 73.4)$
8	$P_{10}(180, 88)$	14.77	31.95	"	$k_1 = (182.1, 75.2)$
9	$P_{11}(180, 67)$	2.52	14.21	"	$k_1 = (181.9, 74.3)$
10	$P_{12}(177, 76)$	5.17	19.70	"	$k_1 = (181.4, 74.5)$

$$K_1 = \{P_1, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$$

$$K_2 = \{P_2, P_3\}$$